

Sec. 3.3 Union and intersection

Definition Let A and B be two sets.

- The **union** of A and B , written $\boxed{}$, is the set of elements in $A \boxed{} B$.
Venn diagram:

- The **intersection** of A and B , written $\boxed{}$, is the set of elements in $A \boxed{} B$.
Venn diagram:

Example 1. Let $A = \{a, b, c\}$ and $B = \{a, x, y, b\}$. Then

1. $A \cup B =$

2. $A \cap B =$

Example 2. For any set A , $A \cup \emptyset = \boxed{}$, $A \cap \emptyset = \boxed{}$.

Exercise 3. Let

$$A = \{x \in \mathbb{Z} : 3 \mid x\}$$

$$B = \{x \in \mathbb{Z} : 5 \mid x\}$$

$$C = \{-30, -4, 0, 1, 5, 9\}$$

$$D = \{-10, -6, 1, 4, 9, 15\}$$

- $A \cap C =$
- $C \cup D =$
- $A \cap B =$
- $B \cap (C \cup D) =$

Properties of union and intersection

Associativity

- $(A_1 \cup A_2) \cup A_3 =$
- $(A_1 \cap A_2) \cap A_3 =$
- $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n =$
- $A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n =$

Distributivity

- $A \cap (B \cup C) =$
- $A \cup (B \cap C) =$

Example 4. For $i \in \mathbb{Z}^+$, define

$$A_i = \{i^0, i^1, i^2\}$$

$$B_i = \{x \in \mathbb{R} : -i \leq x \leq 1/i\}$$

$$C_i = \{x \in \mathbb{R} : -1/i \leq x \leq 1/i\}$$

1. $\bigcap_{i=2}^5 A_i =$

2. $\bigcap_{i=1}^{100} B_i =$

3. $\bigcup_{i=1}^{100} C_i =$

Example 5. For each of the following false equalities, draw a Venn diagram for both sides of the equation. Then, find a counterexample using sets A, B, and C to disprove the equality.

- $A \cap (B \cup C) = (A \cap B) \cup C$

- $A \cup (B \cap C) = (A \cup B) \cap C$